HackAP Hackathon – Agentic AI
Problem Statements

Note: Please read the pages after the problem statements to learn why these problems matter, the key requirements for all the problems and the difficulty levels.

This document outlines the 6 core problem statements for this Agentic AI Hackathon. Each challenge is designed for multidisciplinary teams, requiring a blend of technical execution and strategic business planning.

Problem 1: Autonomous Life Admin Agent

What's the Problem?
People struggle to manage fragmented life tasks (bills, appointments, deadlines, renewals). Current tools remind—but don't act.

What You Need to Design:
Build an agentic system that:

- Understands user intent and responsibilities
- Takes autonomous actions (book, schedule, remind, follow-up)
- Negotiates conflicts (time, priority, cost)
- Continuously learns preferences

Real-World Impact:

- Fully automated personal operations
- Reduced cognitive load
- A step toward "digital executive assistants"

Problem 2: Multi-Agent Startup Builder

What's the Problem?
Early-stage founders struggle to execute across domains (tech, marketing, legal, finance).

What You Need to Design:
Create a multi-agent system where:

- Different agents act as roles (CTO, CMO, CFO)
- Agents collaborate, debate, and make decisions
- System executes tasks (market research, MVP planning, GTM strategy)
- Human acts as high-level supervisor

Real-World Impact:

- Democratized entrepreneurship
- Faster startup execution
- Reduced dependency on large teams

Problem 3: Autonomous Compliance Agent for Businesses

What's the Problem?
Compliance is complex, evolving, and manually intensive.

What You Need to Design:
Build an agent that:

- Reads and interprets regulations
- Maps them to business operations
- Monitors compliance continuously
- Takes corrective actions automatically

Real-World Impact:

- Reduced legal risks
- Automated governance
- Scalable compliance systems

Problem 4: Agentic Market Intelligence Network

What's the Problem?
Businesses rely on static reports instead of continuously evolving market insights.

What You Need to Design:
Create a network of agents that:

- Continuously scan market signals (trends, competitors, sentiment)
- Generate evolving insights
- Recommend strategic actions
- Simulate outcomes

Real-World Impact:

- Real-time business intelligence
- Competitive advantage
- Faster decision-making

Problem 5: Autonomous Skill Acquisition Agent

What's the Problem?
Learning new skills requires structured effort, guidance, and feedback loops.

What You Need to Design:
Build an agent that:

- Identifies required skills for a goal
- Designs a learning path
- Practices, evaluates, and improves
- Demonstrates acquired capability

Real-World Impact:

- Accelerated skill development
- Workforce transformation
- Personalized upskilling

Problem 6: Real-Time Startup Risk Monitoring Agent

What's the Problem?
Investors lack continuous visibility into startup risks after investing.

What You Need to Design:
Create an agent that:

- Monitors signals (performance, market changes, team activity)
- Detects early warning signs
- Predicts potential failures
- Alerts stakeholders proactively

Real-World Impact:

- Reduced investment losses
- Proactive intervention
- Continuous portfolio monitoring

Key Requirements for All Solutions

Design Requirements:

• Platform: Web-based platform (mandatory), mobile optional
• Architecture: Agentic AI system (single or multi-agent architecture required)
• Autonomy: Must demonstrate decision-making, task execution, and iterative improvement
• Human-in-the-Loop: Optional but recommended for supervision and feedback
• Memory: Persistent memory (short-term + long-term context handling)
• Reasoning: Multi-step reasoning and task decomposition capability
• Integration: Ability to integrate APIs, tools, or external systems
• User Interface: Simple dashboard to visualize agent decisions, actions, and outputs
• Transparency: Explainable outputs (why agent made a decision)
• Data Security: Secure handling of user data and agent actions
• Scalability: Modular agent design (extendable with new agents/tools)

Success Criteria:

• Agentic Behavior: Demonstrates true autonomy (not just prompt-response systems)
• Practicality: Solves a real-world problem effectively
• Adaptability: System improves based on feedback or new data
• Reliability: Handles failures, retries, and edge cases
• Explainability: Clearly explains decisions and actions

• Efficiency: Optimizes tasks with minimal human intervention
• Innovation: Goes beyond basic AI usage into system-level thinking
• Impactful: Shows measurable real-world or business impact

Why These Problems Matter:

These problem statements focus on the next evolution of AI — Agentic Systems, where AI moves from:

- Passive tools → Autonomous actors
- Single responses → Continuous execution
- Static logic → Adaptive intelligence

They aim to:
• Enable end-to-end automation of complex workflows
• Build multi-agent ecosystems that collaborate and evolve
• Solve high-friction real-world problems in business, life, and innovation
• Push participants toward building systems, not just models

Difficulty Scores (Scale: 1–10, where 10 is most difficult)

Problem 1: Autonomous Life Admin Agent — Score: 8/10

• Requires real-world task execution and integrations
• Complex scheduling and conflict resolution logic
• Needs adaptive learning of user behavior

Problem 2: Multi-Agent Startup Builder — Score: 9/10

• Multi-agent collaboration and role simulation
• Requires coordination, reasoning, and task execution
• High system design complexity

Problem 3: Autonomous Compliance Agent — Score: 9/10

• Requires understanding and mapping regulations
• Continuous monitoring and dynamic rule application
• High domain + technical complexity

Problem 4: Agentic Market Intelligence Network — Score: 9/10

• Multi-agent data aggregation and analysis
• Requires real-time insights and simulations
• Complex orchestration

Problem 5: Autonomous Skill Acquisition Agent — Score: 9/10

• Long-term planning, execution, and evaluation
• Requires adaptive learning loops
• High system intelligence requirement

Problem 6: Real-Time Startup Risk Monitoring Agent — Score: 8/10

• Continuous monitoring and prediction
• Multi-signal analysis
• Moderate to high complexity

Rationale:

• Scores 1–4: Basic automation (Not applicable for this hackathon)
• Scores 5–7: Intermediate agent systems with limited autonomy
• Scores 8–9: Advanced agentic systems with real-world execution
• Score 10: Frontier-level systems with autonomy, evolution, and multi-agent intelligence

Overall Scoring Framework

Total Points: 50

- Agentic Architecture & Implementation (20 points)
  (Autonomy, multi-agent design, execution capability)

- Problem Understanding & Solution Design (15 points)
  (Clarity, approach, system thinking)

- User Experience & Explainability (8 points)
  (Dashboard, clarity of decisions, usability)

- Innovation & Impact Potential (7 points)
  (Creativity, real-world applicability, uniqueness)